Eq. 10.29 in terms of the position and momentum operators, and the ground-state wave function $\psi_0(x)$:

$$\frac{i}{\sqrt{2\omega\hbar}}(\mathbf{P} - i\omega\mathbf{X})\psi_0(x) = 0,$$

or, dividing by the constant factor,

$$(\mathbf{P} - i\omega\mathbf{X})\psi_0(x) = 0.$$

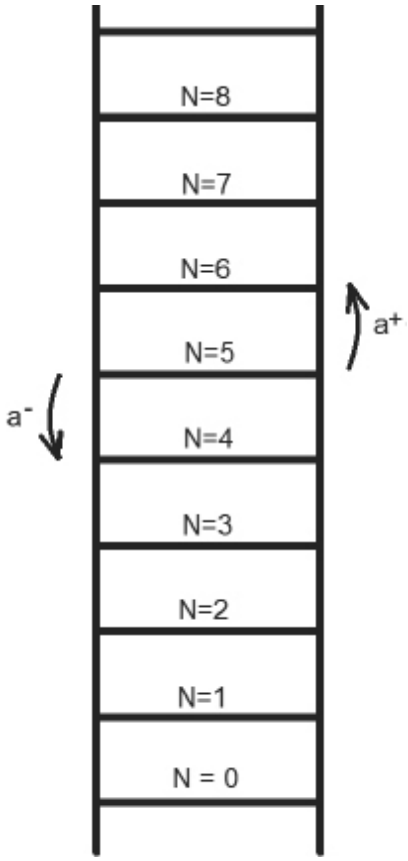


Figure 10.2: Harmonic Oscillator Energy Level Ladder. Energy levels are evenly spaced. $\mathbf{a}^+$ and $\mathbf{a}^-$ raise and lower the energy level respectively. **N** has a lower limit of zero (the ground state), but no upper limit.

If we now replace **P** with $-i\hbar\frac{d}{dx}$, we get a first-order differential equation that is much simpler than the second-order Schrödinger equation: